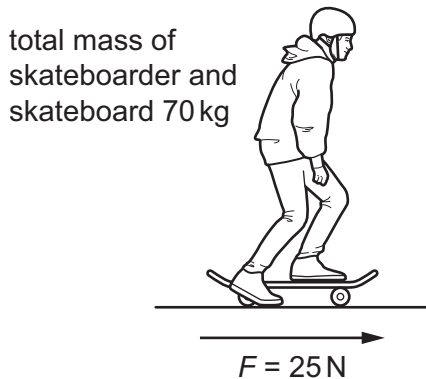


- 16** A skateboarder and her skateboard have a total mass of 70 kg. She pushes on the ground with her foot to create a forward force F of 25 N on herself and the skateboard, as shown in the diagram.



The skateboarder and skateboard travel forwards a distance of 0.50 m before the skateboarder lifts her foot from the ground.

What is the work done by F on the skateboarder and skateboard?

- A** 13 J **B** 50 J **C** 340 J **D** 360 J

- 17** A turbine at a hydroelectric power station is situated at a vertical distance of 20 m below the level